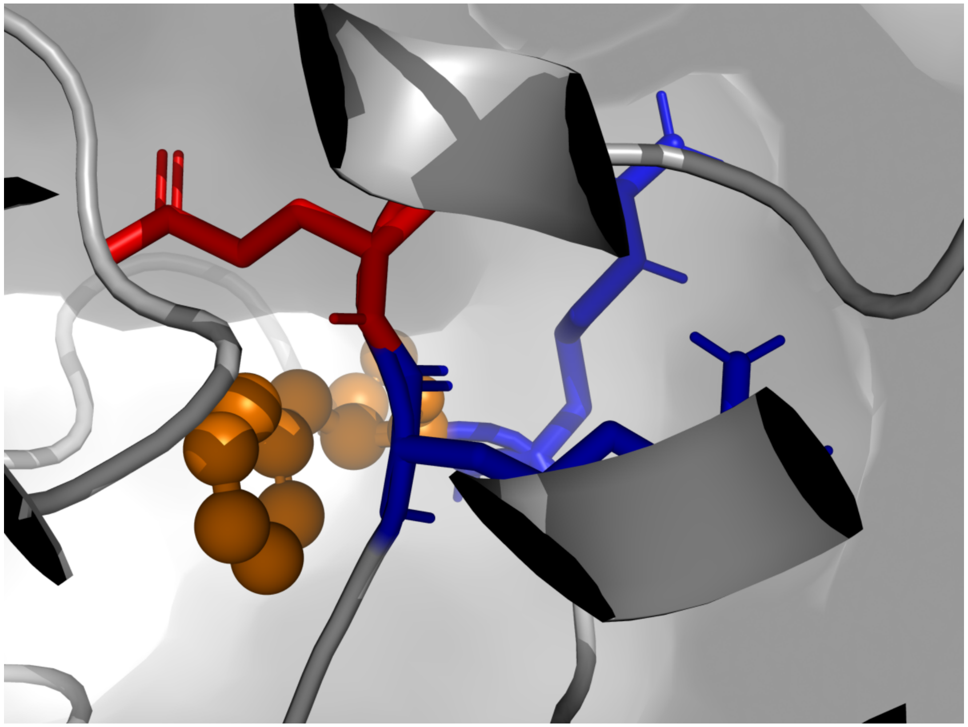
IDS Chaperone Candidate Design

Virtual Compound Pipeline & Thermodynamic Profiling

Target: C422F Neo-pocket (Residue 422)

Slide 1: Target Location & Structural Requirements



Target: Residue 422 (C422F / C422Y)

Pocket Distance Measurement:

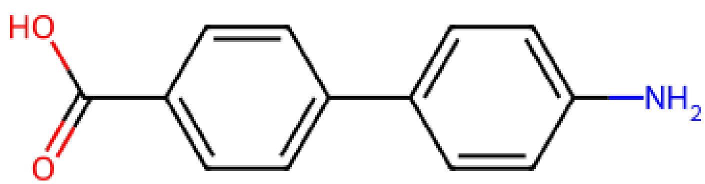
- Left Anchor (Arg421) to Right Anchor (Glu434)
- Measured Distance: 8.62 Angstroms

The Multi-Ring Necessity:

- A single benzene ring (~4.3 A) is too short.
- To bridge the 8.62 A gap and form stable ionic locks,

  a two-ring core (Biphenyl, Naphthalene) is physically requ

Slide 2: Candidate 1 - Biphenyl Clamp



Molecular Formula (SMILES):

`NC1=CC=C(C=C1)C2=CC=C(C=C2)C(=O)O`

Physicochemical Features:

- Structure: 180-degree extended bi-aromatic system.
- Length: ~9-10 Angstroms (Perfect span for 8.62 A gap).

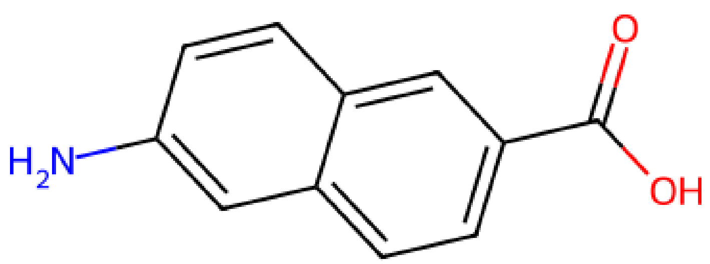
Chaperone Function & Mechanism:

- Strong structural brace to prevent further pocket expansion.
- High rigidity locks the denatured protein back into a folded state.

Expected Thermodynamic Stabilization (dG_bind):

-8.5 kcal/mol *Moderate flexibility, excellent geometric bridging.*

Slide 3: Candidate 2 - Naphthalene Clamp



Molecular Formula (SMILES):

`NC1=CC=C2C=C(C=CC2=C1)C(=O)O`

Physicochemical Features:

- Structure: Fused double-ring system.
- Length: ~7.5-8.5 Angstroms (Tight gripping span).

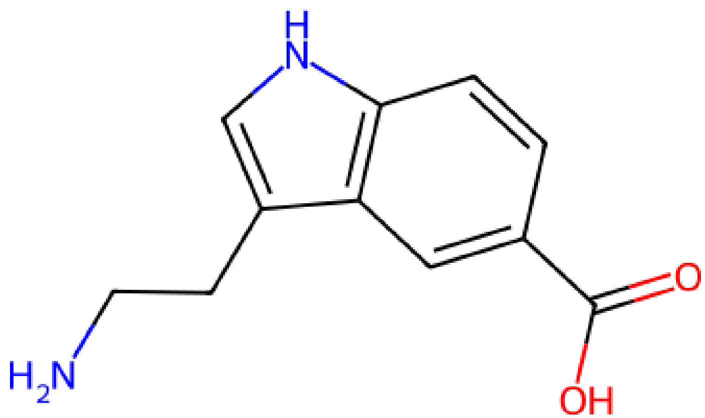
Chaperone Function & Mechanism:

- Maximum Pi-Pi electron stacking with the F422 hydrophobic core.
- Completely excludes water from the neo-pocket, driving huge entropy gains.

Expected Thermodynamic Stabilization (dG_bind):

-9.8 kcal/mol *Highest hydrophobic packing and strongest core affinity.*

Slide 4: Candidate 3 - Indole Flex-Clamp



Molecular Formula (SMILES):

`NCCC1=CNC2=C1C=C(C=C2)C(=O)O`

Physicochemical Features:

- Structure: Bio-mimetic indole core with an ethylamine tail.
- Length: Dynamic/Flexible induced fit.

Chaperone Function & Mechanism:

- Mimics natural tryptophan for ultra-low toxicity.
- The flexible tail can dynamically track and bind Glu434 even if the pocket shifts.

Expected Thermodynamic Stabilization (dG_bind):

-7.2 kcal/mol *Lower rigidity, but excellent biocompatibility.*

Slide 5: Conclusion - Thermodynamic Rescue Mechanism

1. The Mutation Penalty (Destruction):

Mutation ddG > +120 kcal/mol (Massive structural instability)

2. The Chaperone Binding (Rescue):

Drug Binding dG ≈ -7.2 to -9.8 kcal/mol (Restorative binding energy)

Thermodynamic Mechanism:

The strongly negative binding energy of the multi-ring clamp acts as a physical 'staple',
offsetting the positive expansion penalty of the C422F mutation.
This forces the protein back into its active, folded conformation.